



University of Lucknow

Faculty of Engineering s Technology

Department of Computer Science s Engineering



Major Project Title: STREAM

Team Members:

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Project Guide: Er. Vanchana Srivastav

Project Overview: This project focuses on developing an automated deep-learning system to identify safe spacecraft landing sites on Mars. Since the Martian surface contains hazards such as steep slopes, craters, and scattered rocks, manual analysis of landing locations is time-consuming and prone to error. To solve this, the proposed system analyses multimodal remote-sensing data including high-resolution optical images (HiRISE/CTX) and Digital Elevation Models (MOLA/HRSC).

The methodology includes preprocessing satellite data, computing terrain features such as slope and surface roughness from DEMs, and generating safety labels based on terrain characteristics. A UNet / DeepLab segmentation model is then trained to classify each surface area as safe or hazardous for landing. The system produces a visual heatmap overlay showing regions with high landing feasibility.

The expected outcome is a reliable, AI-based solution that assists future Mars missions by automating the landing site selection process, improving accuracy, and reducing human effort. This project demonstrates the application of artificial intelligence in planetary exploration and can be extended to Moon and asteroid landing missions as well.

Technology Used:

The project is developed using Python with deep learning frameworks such as **PyTorch/TensorFlow** for training the model that detects safe landing areas on Mars. Satellite image processing and **Digital Elevation Model (DEM)** analysis are carried out using libraries like **OpenCV, GDAL, and Rasterio**. Additional Python libraries including **NumPy, Pandas, Matplotlib, and Scikit-learn** are used for data handling, visualization, and model evaluation. Experiments and model training are performed in **Jupyter Notebook / Google Colab** using GPU acceleration.

Progress till 30 November 2025:

- learning of Deep Learning Concepts like Perceptron, NLP, forward and backpropagation.
- Conducted initial literature review on Mars surface analysis, landing hazards, and the role of deep learning in planetary landing site prediction.
- Finalized the project title, aim, scope, and expected outcomes.
- Identified the primary datasets to be used, including NASA HiRISE/CTX satellite images and MOLA/HRSC Digital Elevation Models for elevation-based hazard assessment.
- Set up the development environment using Python, Jupyter Notebook/Google Colab, and installed essential libraries such as PyTorch/TensorFlow, NumPy, Pandas, Matplotlib, OpenCV, and GDAL.
- Downloaded sample satellite imagery and DEM tiles for one selected region of Mars for experimentation.

Guide Comments:

Guide Signature: